

Abstract

The aim of this sub-project is to develop an image processing component to support low-cost automated greenhouses. The image processing application must enable the greenhouse system to establish various aspects about the plants from the images taken. These aspects include plant health, plant growth and plant disease identification from the images of different parts of the plants in the greenhouse.

Fill out the introduction.

What is the purpose of this section? To introduce background information about phenotyping in relation to plants. - Why it is useful? - Relations to current study at hand?

Context for plant phenotyping? It is of vital importance in plant breeding to be able to propagate desirable heritable traits from plants through controlled human action [11]. This is especially important in the context of food security where a cultivars resistant to adverse abiotic and biotic stresses affect the final overall crop yield. The literature indicates that plant phenotyping has become an effective tool in plant breeding methods. [1].

Points about plant phenotyping Plant phenotyping is then the evaluation of phenotypes in plants with the main goal of producing quantitative data to study the dynamic interaction between a plant and its environment [3].

Explanation of relevance of plant phenotyping for current study? The term phenotype was first employed by Johannsen [7]. Who stated that a phenotype is all types of organism which are identified from one another by inspection or by some measurement. So it is an observable primitive trait or a complex trait of a plant that can be identified visually or measured and evaluated numerically. Therefore the data produced by plant phenotyping assist in choosing the genetic material (germplasm) that will enable plant breeders to induce traits in cultivars which are optimal for various different growing environments (like water scarce environments).

Why plant phenotyping in the context of this study? Conventional plant phenotyping suffers from scalability issues and is often constrained by costs, time, and human effort required. Furthermore many methods are destructive to the cultivar, making it hard to replicate, and compare measurements of phenotypes over time through the ontogenesis of a cultivar. However in the past decade image processing techniques have been applied successfully to acquire phenotypical data in a non-destructive manner [2]. There exists many different image techniques that have been applied in greenhouses and in the field which allow for the assessment of resistance to disease, biomass, soil salinity, etc using various different sensors, and techniques.

Definition 2: The phenotype can be seen as the combination of all the morphological, physiological, anatomical, chemical, developmental, and behavioral characteristics that, when put together, represent the individual organism. Phenotyping, which is the process of characterizing the phenotype [5]

Definition 3: For the purposes of this review, we refer to phenotyping as the set of methodologies and protocols used to measure plant growth, architecture,

and composition with a certain accuracy and precision at different scales of organization, from organs to canopies. The goal of modern plant phenotyping is to deliver quantitative data on the dynamic responses of plants to the environment. [3]

Definition 4: The broad evaluation of complex plant traits like growth, development, tolerance, resistance, architecture, physiology, ecology, and yield as well as the fundamental measurement of individual quantitative parameters that serve as the foundation for more complex traits is known as plant phenotyping [10]

Definition 5: Plant phenotyping is the comprehensive assessment of complex plant traits such as growth, development, tolerance, resistance, architecture, physiology, ecology, yield, and the basic measurement of individual quantitative parameters that form the basis for more complex traits [8]

What is the purpose of this section?

- Discuss various image processing techniques in plant phenomics
- Discuss various phenotypical data that can be obtained via images

What are the main points?

- Use principles of electromagnetic radiation + spectroscopy
- Fluorescence, thermal imaging, hyper spectral, tomographic etc
- Root methods not well developed
- Machine learning techniques often also applied as well

Why these points are important to main topic/paragraph?

- Various image techniques can be employed to give us insight into plant phenotypical traits
- These insights can be applied in plant breeding to guide agriculture.

TODO List

- The imaging sensing techniques and what they can possibly detect is important. For example soil salinity can cause water stress, which can affect plant metabolism, which in turn can affect growth. On the other hand soil salinity can also cause diseases. So its hard to map one parameter to under one sensor technology. Multi sensor approach ideal. Best to discuss the image processing techniques in terms of the sensing technology, what they can measure, and what phenotype that helps analyse. Give examples in green houses.

Image processes that have been applied and discussed in the literature focus on analyzing and capturing the spectral reflectance properties of leaves, plant and canopies, thermal temperature, morphology of plant, and root system structures [13]. The main problem before image processing can be applied is image acquisition which depends on what sensors and measurement technique are used. Therefore current image processing techniques in plant phenotyping is closely related to the sensors and measurement techniques used in image acquisition [12].

What affects plant growth?

Namely we care about yield and biomass which is restricted by production conditions under stress, and disturbances. Which may be abiotic or biotic. [4].

Stomotal Conductance Thermal imaging

A key phenotypical parameter to measure in these contexts is stomotal conductance. Plants are cooled by transpiration the more gas that is diffused through their stomota, when a plant has inadequate amounts of water its temperature rises as transpiration decreases as a stress response to water scarcity [9]. Thermal imaging can be used to detect this [6].

Fill out the applications in low cost greenhousing techniques section.

Fill out the conclusion section.

References

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